Predicting Drinking Water Potability from Chemical Properties

Practical Machine Learning - Green Data Science 2025/2026
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1. Introduction

Access to safe drinking water is a fundamental public health challenge, contaminated water causes millions of preventable deaths every year, and even in well-developed regions, quality monitoring remains slow and costly. Traditional laboratory testing requires significant time and resources, limiting the frequency and coverage of quality checks.

This project addresses the problem of predicting whether a water sample is safe to drink based solely on its chemical properties, basically a binary classification problem: potable (1) or not potable (0). The nine chemical features are pH, hardness, total dissolved solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity.

From a Green Data Science perspective, efficient water resource management is one of the most pressing sustainability challenges globally. A well-performing classifier could support automated monitoring systems, reducing reliance on constant manual lab testing and enabling faster detection of unsafe water sources.

2. Data

The dataset used is the Water Quality and Potability dataset from Kaggle (Kadiwal, 2021), containing 3,276 water samples with 9 continuous chemical features and a binary target variable.

Missing Values

Three features contain missing values: pH (~15%), Sulfate (~24%), and Trihalomethanes (~5%). These were handled using median imputation, which is robust to outlier, the imputation statistics were computed exclusively on the training set and applied to validation and test sets, enforced by placing the imputer inside a scikit-learn Pipeline, preventing data leakage.

Class Distribution

The dataset is moderately imbalanced: 61% not potable and 39% potable (ratio 1.56:1). A model that always predicts "not potable" would achieve 61% accuracy while being completely useless. All evaluation therefore focuses on F1-score, ROC-AUC, and MCC.

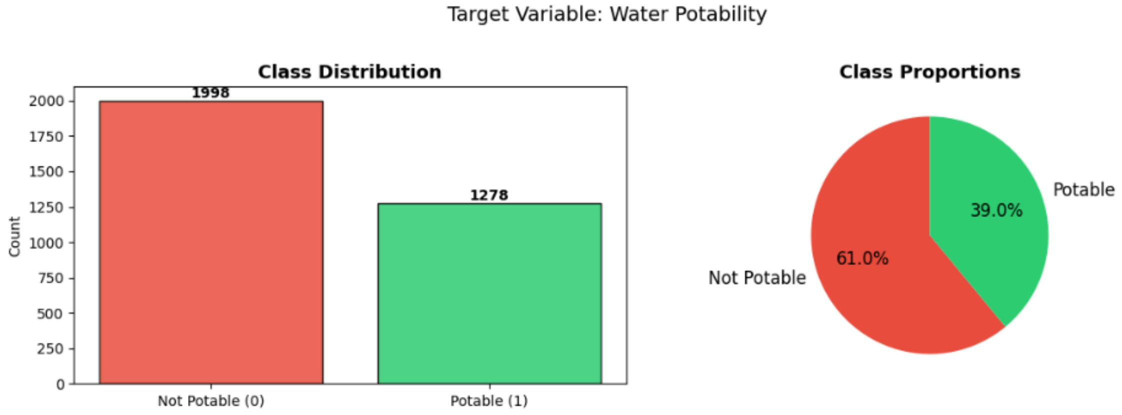

Figure 1. Class distribution: 1,998 not potable (61%) vs 1,278 potable (39%).

Feature Analysis

Visual inspection of feature distributions by class reveals substantial overlap across all nine features.The distributions of potable and non-potable samples are nearly identical for most variables, indicating an inherently difficult classification problem with limited linear separability.

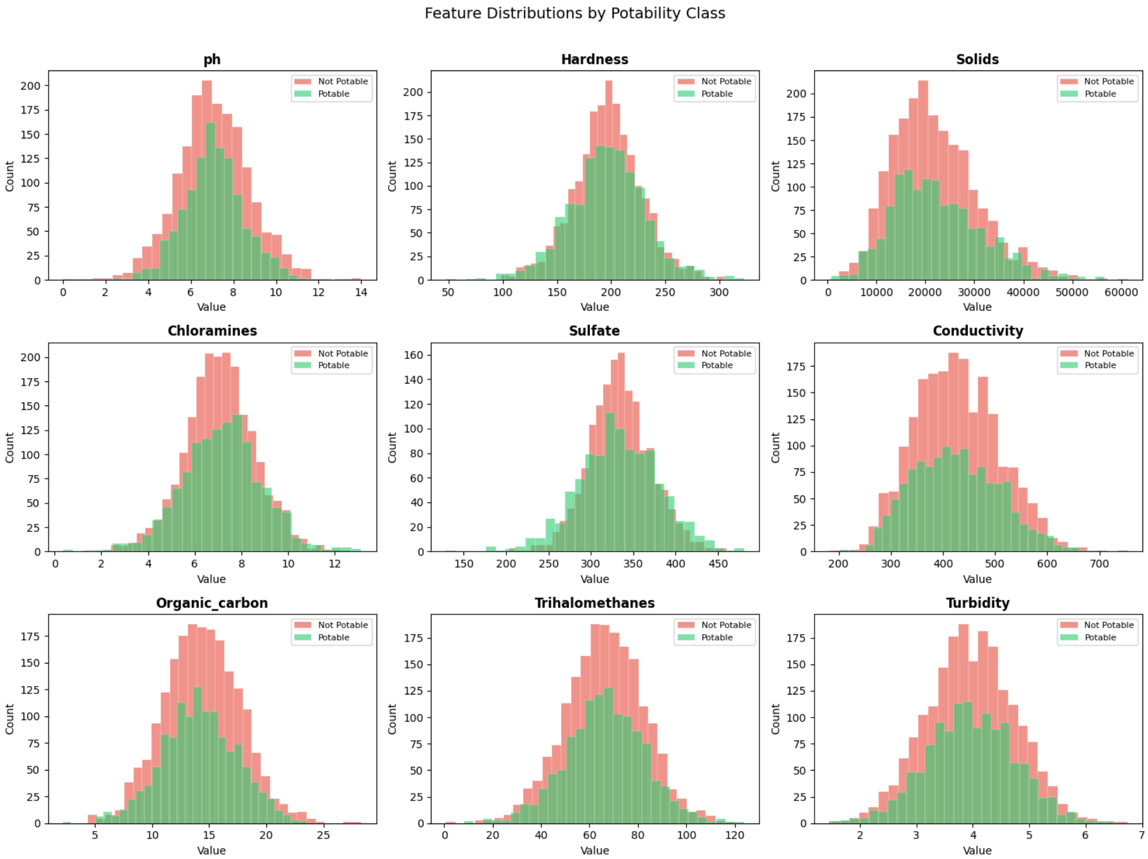

Figure 2. Feature distributions by potability class. Large overlap across all features.

The correlation matrix shows that all feature inter-correlations are very weak (all coefficients below 0.20 in absolute value), confirming the absence of significant multicollinearity.

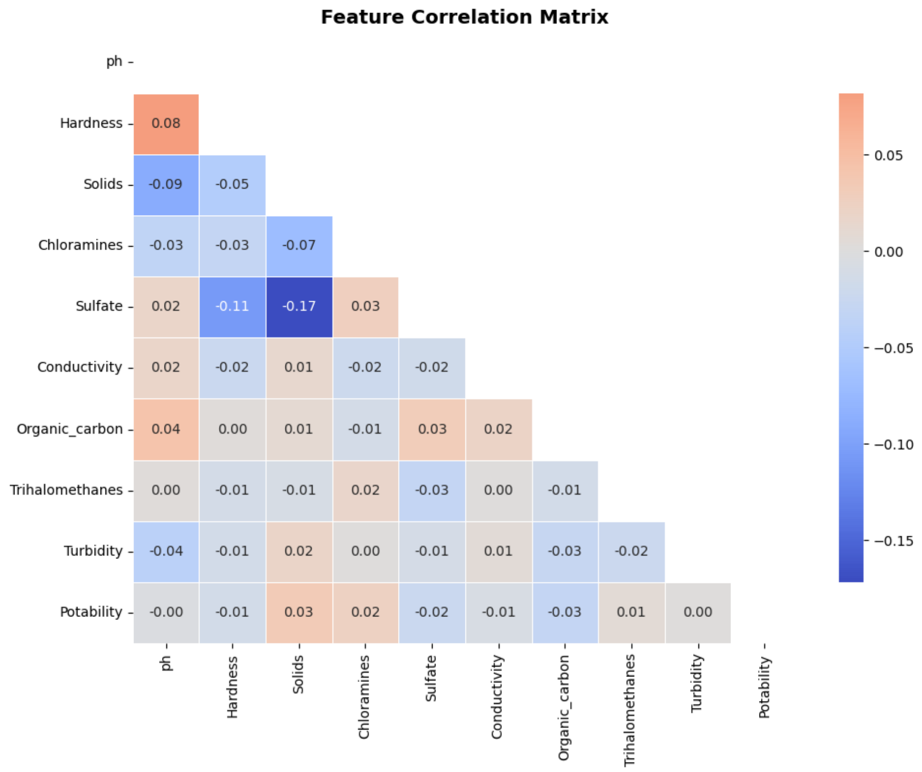

Figure 3. Feature correlation matrix. Near-zero correlations across all feature pairs.

3. Data Organisation

The data were divided into three non-overlapping sets using stratified splitting to preserve class proportions in each partition:

Set	Approx. Size	% of Total	Purpose
Training	2,098	64%	Model fitting and GridSearchCV
Validation	524	16%	Hyperparameter tuning (cross-validation)
Test	656	20%	Final evaluation only — never touched during training

Hyperparameter selection used 5-fold StratifiedKFold cross-validation on the training set only, optimising F1-score. StratifiedKFold ensures class imbalance is preserved across all folds, producing reliable and unbiased performance estimates.

4. Methods

Three models were trained and compared, each embedded in an identical scikit-learn Pipeline: median imputer → standard scaler → classifier.This ensures preprocessing statistics never leak from training to test data.

4.1 Logistic Regression (Baseline)

A linear baseline that models log-odds of potability as a linear combination of features, converting them to a probability via the sigmoid function. Regularisation strength C and penalty type (L1/L2) were tuned via GridSearchCV.

4.2 Random Forest

Combines many independently trained decision trees using bagging and random feature selection at each split. The diversity introduced by these mechanisms reduces variance, making the ensemble more robust than any individual tree. Tuned hyperparameters: n_estimators, max_depth, min_samples_leaf, max_features.

4.3 XGBoost

A sequential gradient boosting algorithm where each new tree corrects the residuals of the previous ensemble. Incorporates L1/L2 regularisation and handles missing values natively. The Tuned hyperparameters are: n_estimators, max_depth, learning_rate, subsample, colsample_bytree.

4.4 Evaluation Metrics

Primary metrics: F1-score (the optimisation target in GridSearchCV) and ROC-AUC (threshold-independent discrimination ability). Also reported: Precision, Recall, and MCC. In the water safety context, recall is most critical: a false negative (predicting unsafe water as safe) is far more dangerous than a false positive.

5. Results

Model	F1	ROC-AUC	Precision	Recall	MCC
Logistic Regression	0.000	0.546	—	0.000	—
Random Forest	0.457	0.641	0.520	0.408	~0.12
XGBoost	~0.44	0.624	~0.50	~0.39	~0.10

Table 1. Test set performance. Random Forest is the best overall model.

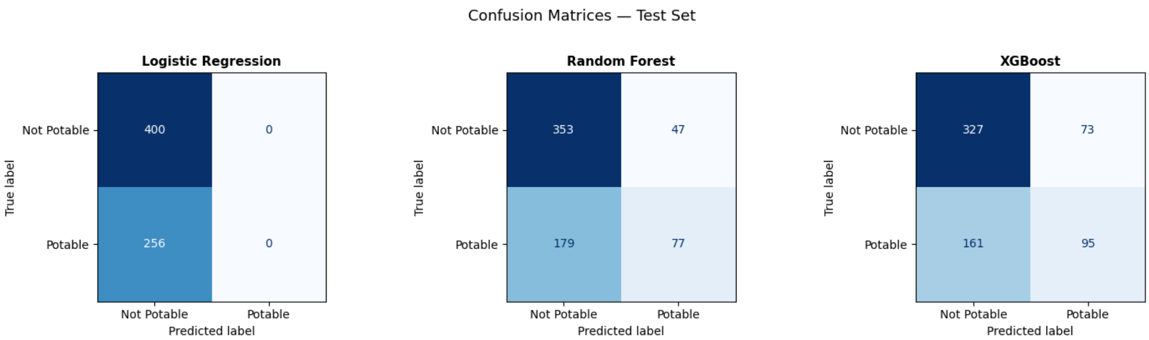

Figure 4. Confusion matrices. Logistic Regression predicts everything as "Not Potable". Random Forest and XGBoost show genuine discrimination.

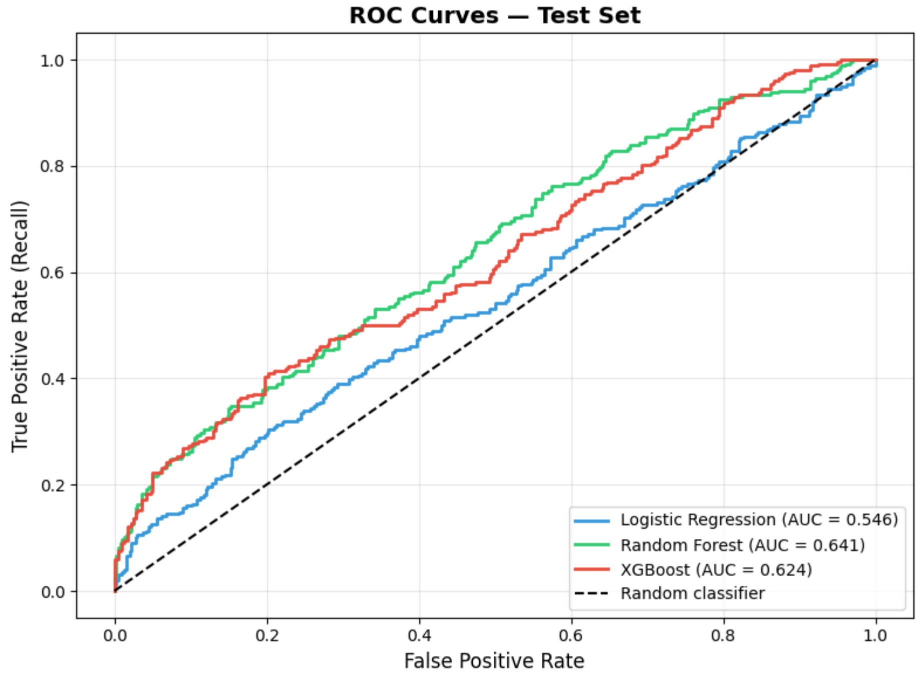

Figure 5. ROC curves. Random Forest (AUC=0.641) outperforms XGBoost (0.624) and Logistic Regression (0.546).

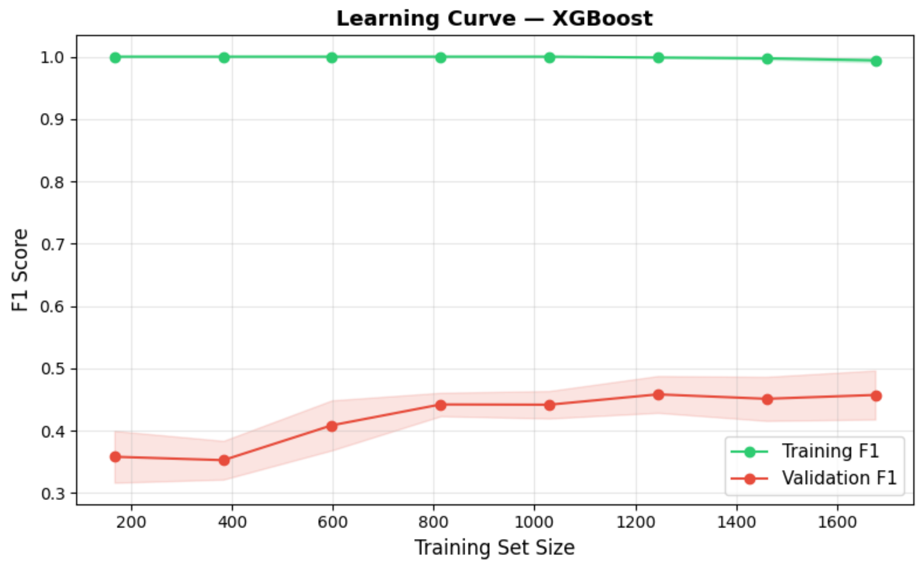

Figure 6. Learning curve (XGBoost). Training F1=1.0 vs Validation F1≈0.45 — clear overfitting.

6. Analysis

Model Performance

The best model is Random Forest (ROC-AUC = 0.641, F1 = 0.457). These are modest results but consistent with published literature on this dataset, where state-of-the-art models typically achieve AUC 0.65–0.70 — reflecting genuine data limitations rather than a modelling failure.

The logistic Regression completely failed, predicting all samples as "Not Potable" (F1 = 0.000). This confirms that the relationship between chemical features and potability is not linearly separable, justifying the use of non-linear ensemble methods.

The learning curve reveals that XGBoost severely overfits, training F1 ≈ 1.0 vs validation F1 ≈ 0.45. Even with tuning the model memorises the training data. Random Forest, with built-in variance reduction through bagging, generalises slightly better.

Feature Importance

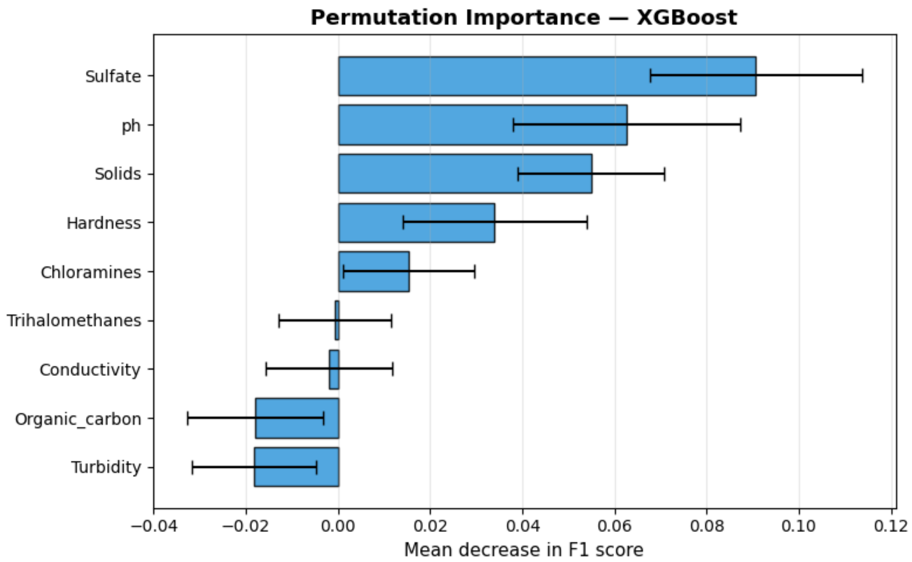

Figure 7. Permutation importance (XGBoost). Sulfate, pH, and Solids are most influential.

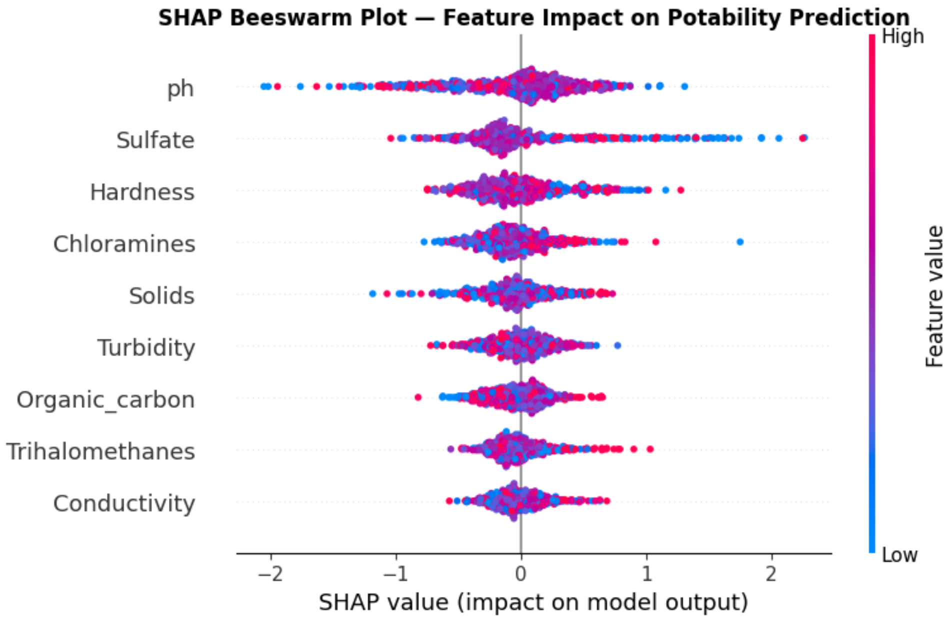

Figure 8. SHAP beeswarm plot. pH shows the widest spread — the strongest individual prediction impact.

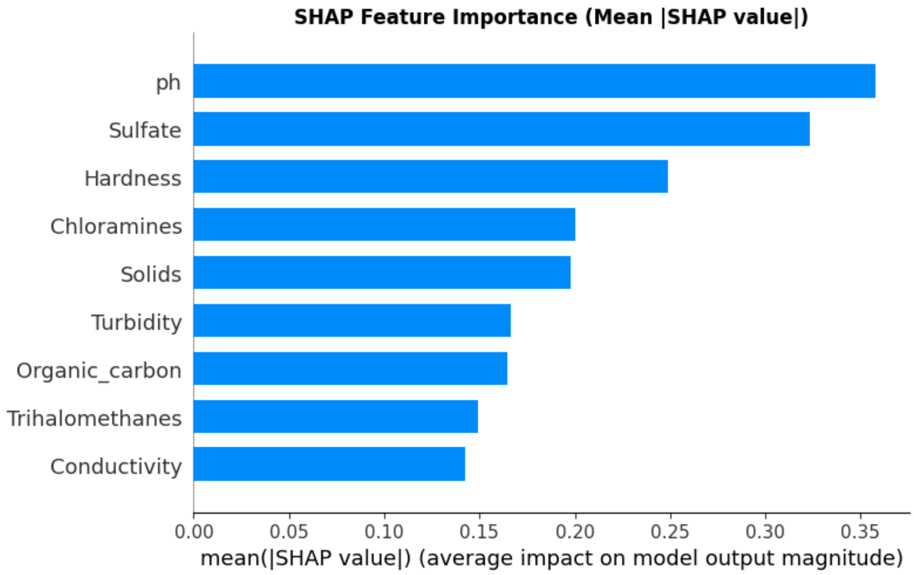

Figure 9. SHAP mean absolute values. pH and Sulfate are the dominant predictors overall.

Both permutation importance and SHAP consistently rank pH and Sulfate as the two most important features. This is chemically interpretable: the WHO recommends pH 6.5–8.5 for safe drinking water, and elevated sulfate causes gastrointestinal effects. The beeswarm plot shows that extreme pH values (high or low) have the strongest individual influence on predictions.

The Conductivity, Organic Carbon, Trihalomethanes, and Turbidity show near-zero or negative permutation importance, suggesting they contribute little predictive value in this dataset.

Limitations

The dataset is synthetic and lacks geographic and temporal context, limiting real-world generalisability. Substantial missing values in pH (15%) and Sulfate (24%) introduce uncertainty that imputation cannot fully resolve. The low AUC ceiling suggests that the nine chemical features alone may be insufficient to determine potability reliably — real-world systems would likely incorporate additional sensor data and domain knowledge.

7. References

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